

THE HONG KONG POLYTECHNIC UNIVERSITY
DEPARTMENT OF COMPUTING
EXAMINATION

Course : Broad Discipline of COMP-61431, BSc Financial Technology-61434,
BSc Investment Science & Finance Analytics-63426,
BSc Data Science & Analytics-63425

Subject : COMP4431 Artificial Intelligence

Group : 1011, 181

Session : 2022 / 2023 Semester I

Date : 02 December 2022

Time : 12:30 - 14:30

Time Allowed: 2 Hours

Subject Lecturer: Dr LIU Fiona

This question paper has 4 pages.
(Some pages may be intentionally omitted.)

Instructions to Candidates:

Answer ALL the questions on the answer book.
Close book.
Calculator is allowed.

Do not turn this page until you are told to do so!

Section B**[80 Points]**

Question 2. Answer the question in (a) short paragraph(s).

2.1. Could you please list the seven research directions of artificial intelligence when it was formally proposed at Dartmouth Colleague in 1956? **[7]**

2.2. Could you give an example to illustrate how the strategy of least constraining value improve the efficiency of the constraint satisfaction problem solving? Could you give an example to illustrate how the strategy of most constrained variable improve the efficiency of the constraint satisfaction problem solving? **[4]**

2.3. Could you give an example under multiagent environment to illustrate how α - β pruning improves efficiency of problem solving? **[3]**

2.4. Could you give an example, including both the knowledge base and the query procedure to show the difference between forward chaining and backward chaining of knowledge-based agent? **[6]**

Question 3. Solve the eight puzzle problems using search algorithms. The eight puzzle is a sliding puzzle having eight square tiles numbered 1–8 in a frame that is 3 tiles high and 3 tiles wide, leaving one unoccupied tile position. Tiles in the same row or column of the open position can be moved by sliding them horizontally or vertically, respectively. The goal of the puzzle is to place the tiles in numerical order as shown in Figure 1(a).

	1	2
3	4	5
6	7	8

(a)

1	4	2
3	5	
6	7	8

(b)

	7	2
5	4	3
1	6	8

(c)

Figure 1. Figure of question 3

3.1. Could you solve the eight-puzzle problem using breadth-first search when the start state is shown in Figure 1(b) and the goal state is shown in Figure 1 (a)? Please write down the expansion of the search tree step by step with the clear order until the solution is found. **[12]**

3.2. If the start state is shown in Figure 1(c), could you illustrate the domination of the heuristics by comparing two heuristic functions? Please write down the estimate cost to goal (as shown in Figure 1 (a)) from the start state (as shown in Figure 1 (c)). **[12]**

Question 4. Answer the questions of neural networks.

4.1. Eight 2-dimensional data points are given as above.

X1	X2	X3	X4	X5	X6	X7	X8
2	-2	1	1	2	1	-1	-2
2	-2	2	2	1	1	-1	-2

The corresponding labels y_i are

Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8
1	1	-1	-1	-1	-1	1	1

The learning rate η is set to 0.1. Please calculate the value of w_i according to the given perceptron learning algorithm. [8]

Perceptron Learning algorithm

1. Initialize a random w_0 as (1,1)
2. For each (x_i, y_i) , η is the learning rate.
 - 2a: calculate the estimated label $y_i^{est} = \text{sgn}(w_{i-1}^T x_i)$
 - 2b: update the weight $w_i = w_{i-1} + \eta(y_i - y_i^{est})x_i$

4.2. Could you please calculate the convolutional result for the image block of 4*4 with the convolutional kernel of 3*3. [4]

100	120	110	110
110	150	140	120
120	140	150	110
110	140	140	120

©

1	1	1
1	1	1
1	1	1

Question 5. Given the data below, decision tree will be constructed to predict the label of yes or no. Could you construct the decision tree to determine the label of the data based on the calculation of information gain and entropy? [12]

Data	F1	F2	F3	F4	F5	Label
1	F1V2	F2V1	F3V1	F4V1	F5V2	YES
2	F1V2	F2V2	F3V1	F4V1	F5V2	NO
3	F1V2	F2V2	F3V1	F4V1	F5V2	NO
4	F1V2	F2V1	F3V1	F4V3	F5V2	NO
5	F1V1	F2V2	F3V1	F4V2	F5V2	YES
6	F1V1	F2V2	F3V1	F4V2	F5V2	YES
7	F1V1	F2V1	F3V1	F4V1	F5V2	YES
8	F1V1	F2V2	F3V1	F4V1	F5V2	NO
9	F1V2	F2V2	F3V2	F4V3	F5V2	NO
10	F1V2	F2V2	F3V2	F4V3	F5V2	NO
11	F1V1	F2V2	F3V2	F4V2	F5V1	YES
12	F1V2	F2V1	F3V2	F4V3	F5V1	NO
13	F1V1	F2V1	F3V2	F4V3	F5V1	YES
14	F1V1	F2V1	F3V2	F4V2	F5V1	YES
15	F1V1	F2V1	F3V2	F4V2	F5V1	YES
16	F1V1	F2V2	F3V2	F4V1	F5V1	NO

Question 6. According to the Bayesian Networks shown in Figure 2, could you please calculate $P(B|C)$. [12]

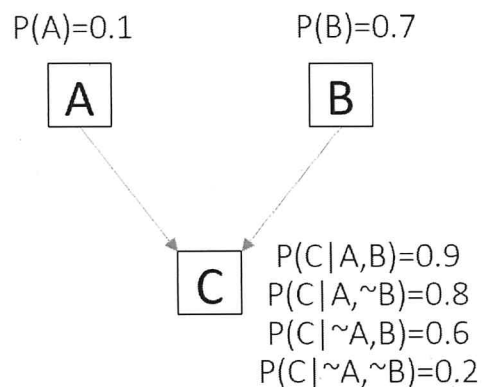


Figure 2. Figure of question 6

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